

# GeoSenEsm

Application User Guide



*Geolocation and Sensor data supported  
Experience Sampling Method*

Poznań, 7th August 2026

This document describes the features and capabilities of the GeoSenEsm application. GeoSenEsm enables the creation of periodic survey forms for respondents in studies that use the Experience Sampling Method (ESM). Respondents' answers are additionally correlated with their location and with sensor readings that measure conditions in their immediate surroundings or on their body. The project ships a mobile application for Android and iOS. To operate a study, researchers also need to host an administrator application for configuring the study, creating surveys and monitoring results.

GeoSenEsm — an abbreviation of *Geolocation and Sensor data supported Experience Sampling Method* — integrates geolocation and sensor data with ESM, a widely used technique that captures participants' real-time self-reports on experiences, behaviours and emotions. By pairing subjective responses with objective environmental metrics, GeoSenEsm helps urban researchers uncover more nuanced relationships between urban environments and human behaviour.

This edition of the guide covers GeoSenEsm 3.0.0, which reorganised the admin panel around a redesigned navigation menu, introduced study-wide *Survey Settings* (study logo, sending-policy calendar visibility, CSV export format), replaced the fixed Xiaomi/Kestrel sensor choice with a configurable Bluetooth Low Energy sensor-profile engine that supports any compatible device without an app update, added per-respondent sensor assignment, per-slot notification rules, an Issues overview for spotting incomplete participation, and timezone-aware scheduling for studies that span multiple time zones.

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# 1 GeoSenEsm project

GeoSenEsm is open-source software distributed under the MIT licence. The full source code lives on GitHub at <https://github.com/geosenesm>.

## 1.1 Project highlights

**Simple, inclusive design.** The main screen is dedicated to surveys only. One question is shown per page and font sizes follow the device's accessibility settings.

**Data ownership and privacy.** All data is stored on a server operated by the researcher. Accounts cannot be self-registered, so the researcher retains full control over who joins the study.

**Scheduling and notifications.** Configure start/finish times per survey and repeat them multiple times a day. Each survey defines its own set of reminder rules (e.g. "15 minutes before the slot closes"), and schedules are interpreted against each respondent's own local time zone.

**Offline first.** Surveys, answers, GPS points and sensor readings are stored locally when the network is unavailable and pushed to the server automatically once connectivity returns.

**Admin portal.** Configure the study, create surveys, manage sending policies, inspect results, and export CSV, JSON or ZIP from any modern browser.

**Location and configurable sensor data.** Enrich responses with GPS and with readings from any Bluetooth Low Energy sensor the researcher has configured — temperature/humidity sensors, pulse oximeters, plant-care sensors and more — without rebuilding the mobile app. Optionally constrain the study to a specific polygon.

**Question types and conditional routes.** Seven question types, section-level visibility rules (always, by group, conditional) and initial-survey grouping.

**Respondent transparency.** Participants can inspect their own submitted locations, live sensor readings and progress calendar inside the mobile app.

## 1.2 System architecture

GeoSenEsm is a multi-platform system based on the Flutter framework for the mobile client and on a Spring Boot / Angular pair on the server. Researchers provision their own server instance; the mobile app then connects to it using a URL supplied by the study administrator.

The server instance consists of a Java API, an admin web application, a relational database (SQL Server, storing the operational data) and a document store (MongoDB, mirroring full response documents for the download tab).



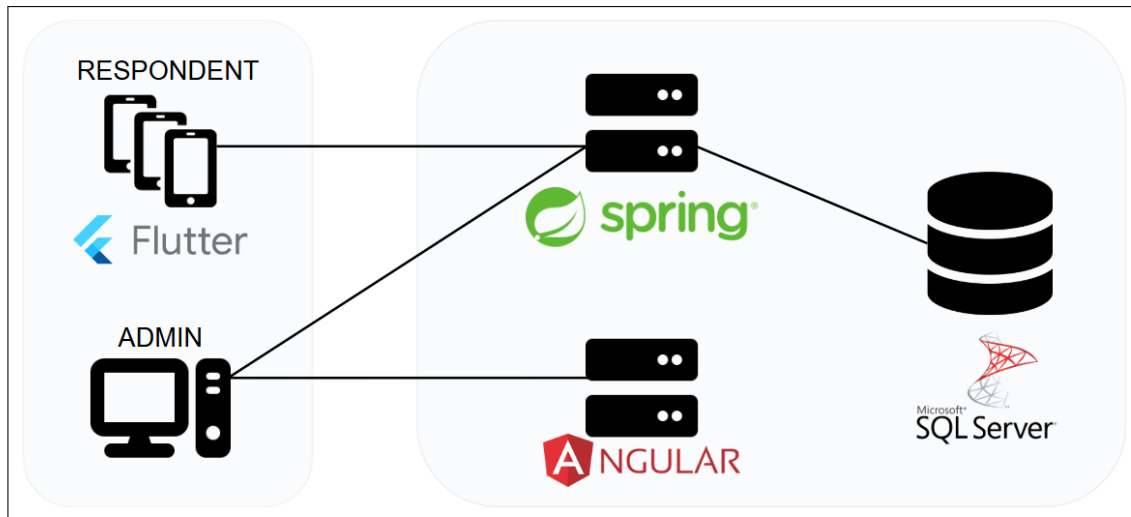


Figure 1.1: *GeoSenEsm system architecture.*

## 1.3 Server deployment

To set up a server, researchers need Docker installed (<https://www.docker.com/>). Docker packages applications together with their dependencies into containers; a single `docker-compose.yml` file is enough to bring up every component of GeoSenEsm without requiring specialist system-administration expertise.

Components deployed as Docker containers are:

1. the Java (Spring Boot) API,
2. the Angular admin panel,
3. SQL Server (operational database),
4. MongoDB (response-document store).

The full deployment definition and step-by-step instructions live in the DevOps repository at <https://github.com/GeoSenEsm/devops>. The only component that is not covered by `docker-compose` is the DNS / reverse-proxy configuration that maps a public domain to the API and admin containers; consult your network administrator if you need help with that step.

## 1.4 Admin panel navigation

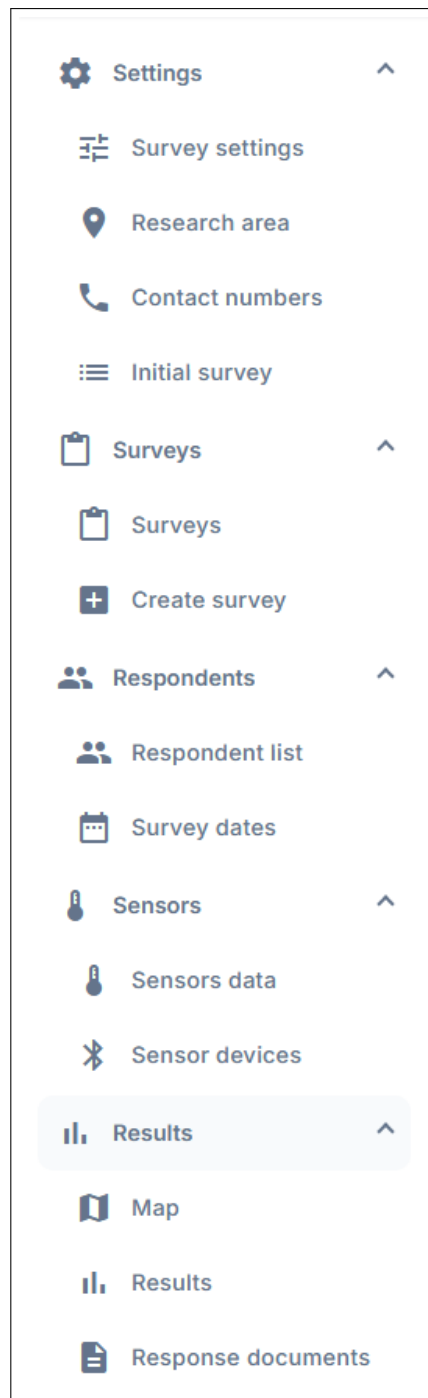
The admin panel is a browser-based application hosted next to the API. Its left-hand navigation (Figure 1.2) groups every screen into six categories, which in turn map onto the next four chapters of this guide:

**Settings** — study-wide configuration; covered in Chapter 2, [Survey Settings](#).

**Sensors** (plus the Sensor Profiles and Integrations pages reached from Survey Settings) — covered in Chapter 3, [Choosing sensors and sensor data](#).

**Surveys and Respondents** — covered in Chapter 4, [Setting up survey](#).

**Results and Statistics** — covered in Chapter 5, [Monitoring and Survey results](#).



**Figure 1.2:** *Left-hand navigation of the admin panel, grouped into Settings, Surveys, Respondents, Sensors, Results and Statistics.*

Researchers open the admin panel in any modern browser at the URL of their deployed instance.

## 2 Survey Settings

This chapter covers the *Settings* navigation group: study-wide options that apply to every survey and every respondent, the research area, the contact numbers shown to respondents, and the mandatory initial survey.

### 2.1 General settings

The *Survey settings* page (Figure 2.1) is new in v3.0.0 and centralises options that used to have no dedicated home:

**Study logo.** An optional JPG or PNG (up to 1 MB) uploaded via *Upload logo* and removed with the corresponding *Remove* action. When set, it is shown at the top of the mobile app’s home screen, reinforcing the study’s identity for respondents.

**Show sending policy calendar.** A toggle that controls whether respondents can open a calendar of scheduled survey slots in the mobile app. Disable it for studies where the researcher would rather not reveal the exact sending schedule to participants.

**CSV format.** The column and decimal separators (e.g. comma vs. semicolon, dot vs. comma) used consistently for every CSV export in the admin panel and for the CSV imports used by the research-area polygon and the sensor-devices whitelist. Studies run from locales where the default comma/dot convention is unusual can switch this once, instead of reformatting every exported file by hand.

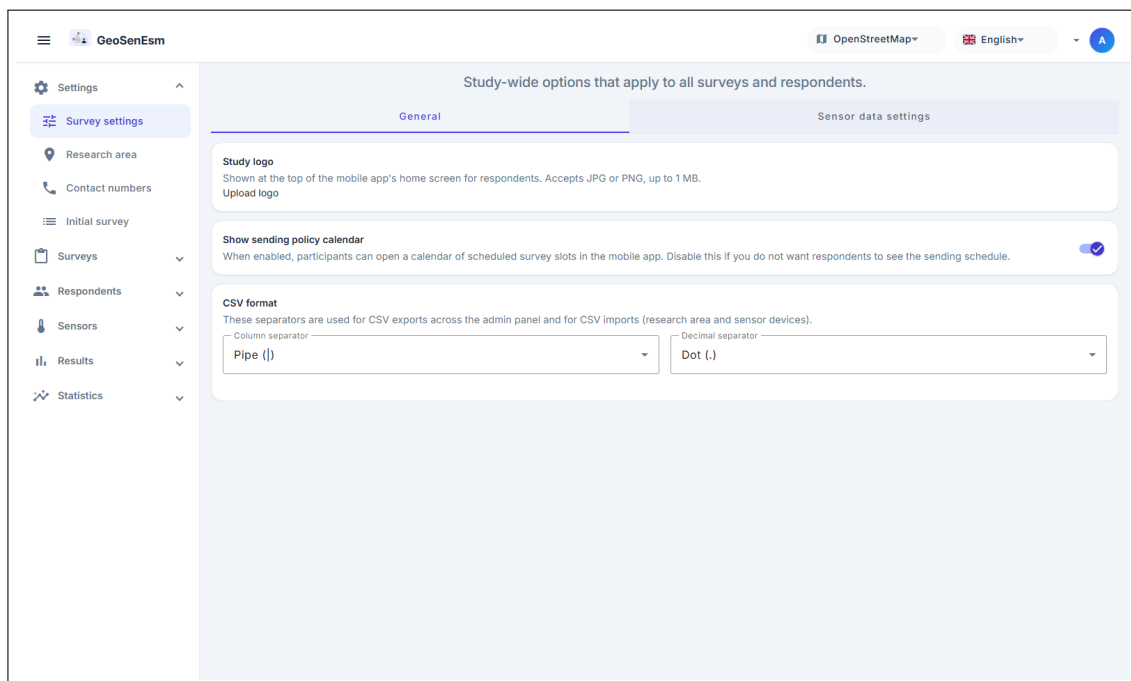


Figure 2.1: *Survey settings* — *General tab*.

The second tab on this page, *Sensor data*, configures which sensor types are collected study-wide; it is described in Section 3.4 of the next chapter, since it is tightly coupled to sensor-profile management.

## 2.2 Research area

Researchers can define an optional polygon representing the geographic scope of the study. Every GPS point that arrives from a respondent is compared against this polygon and marked either *inside* or *outside*. The result is used by the map view, by the *Surveys filled outside research area* chart in Statistics (Section 5.5.5) and by the `outsideResearchArea` filter on the GPS-data export.

The polygon is uploaded as a CSV file with latitude / longitude columns, formatted using the column separator configured in Section 2.1. Points must be counter-clockwise and the polygon must be closed (first and last point identical, minimum three unique points).

|   | A                  | B        |
|---|--------------------|----------|
| 1 | longitude          | latitude |
| 2 | 16.811,52.47382065 |          |
| 3 | 17.038,52.48762081 |          |
| 4 | 17.015,52.35277971 |          |
| 5 | 16.785,52.3603283  |          |
| 6 | 16.811,52.47382065 |          |

Figure 2.2: Example research-area CSV file.

If the research area changes mid-study, existing GPS rows are re-classified automatically.

## 2.3 Contact numbers

Phone numbers registered here are shown to the respondent in the mobile app so they can reach the study team (Figure 2.3).

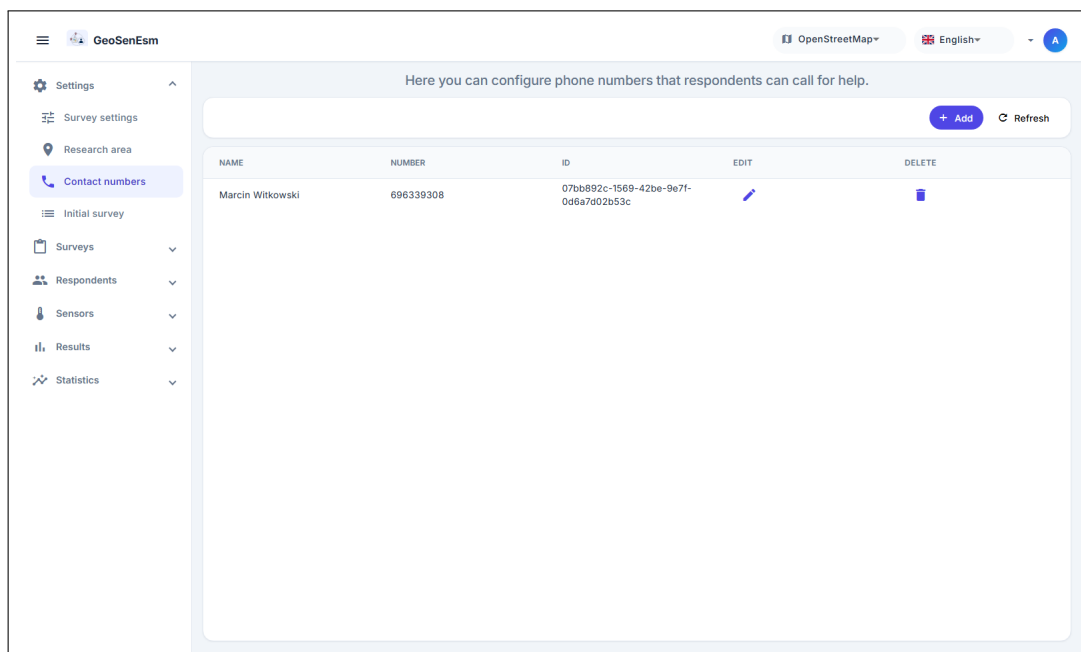


Figure 2.3: Managing study-contact phone numbers.

## 2.4 Initial survey

The initial survey is shown to the respondent on first login and is mandatory (Figure 2.4). Every initial-survey question must be single-choice — the selected option later becomes a column in the respondent record and can be used to route conditional or group-restricted sections in normal surveys.

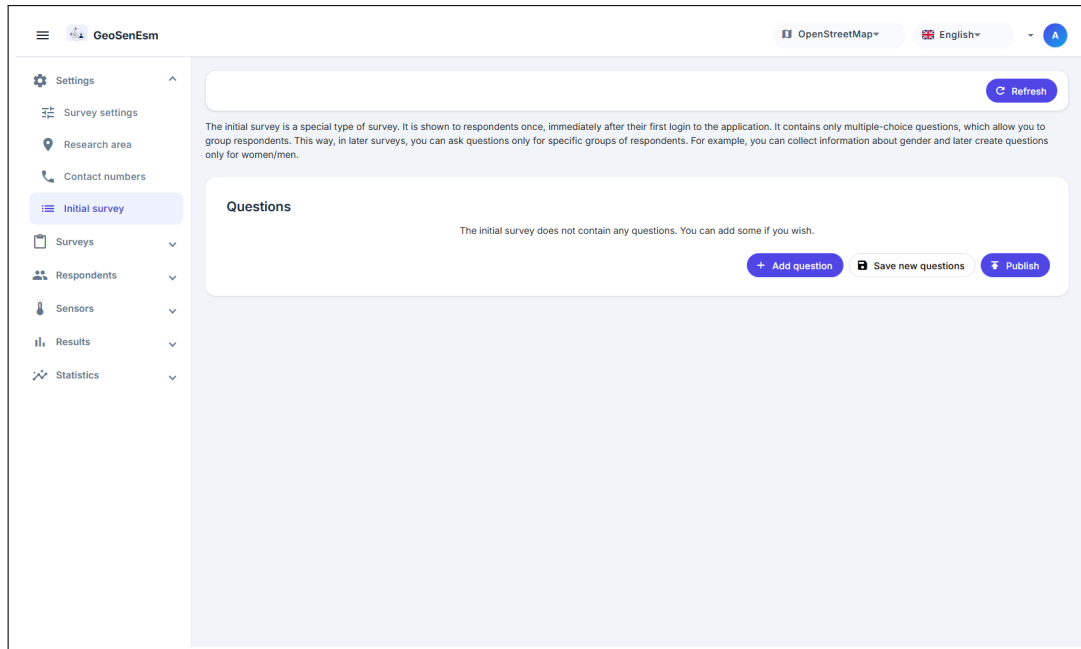


Figure 2.4: Initial-survey editor.

### Sensor and survey setup lock

Once the initial survey is published, the study is considered live. From that point on, sensor-type management, GATT-profile edits and the sensor-data mode toggle described in Chapter 3 are locked to protect data consistency; only per-respondent sensor *assignments* stay editable, since which physical device a respondent carries can keep changing throughout the study.

## 3 Choosing sensors and sensor data

GeoSenEsm 3.0.0 replaces the old hard-coded choice between “Xiaomi”, “Kestrel” or “no sensor” with a configurable Bluetooth Low Energy sensor-profile engine. Researchers pick which physical sensors a study uses and which readings count as “sensor data” entirely from the admin panel; the mobile app downloads the active configuration and knows how to talk to any supported device without a new app release.

### 3.1 Supported sensor types

Seven sensor types ship as ready-to-use templates:

**Xiaomi Mi LYWSD03MMC** — temperature and humidity.

**Kestrel Drop D2** — temperature and humidity, with a unique serial number so no MAC whitelist is required.

**Inkbird IBS-TH1 / Mini / Plus** — temperature and humidity.

**PC-60FW** — pulse oximeter (SpO<sub>2</sub>, pulse rate, perfusion index).

**Bluetooth SIG PLX** — standards-based pulse oximeter.

**Flower Care** — plant sensor (temperature, light, moisture, conductivity).

**Xiaomi Door Sensor 2** — open/closed contact sensor.



(a) Xiaomi Mi LYWSD03MMC.



(b) Kestrel Drop D2.

Figure 3.1: Two of the built-in sensor templates.

Every template describes its Bluetooth discovery rule, its read/notify sequence and how raw bytes decode into values as a small JSON *GATT profile*. Because the mobile app’s Bluetooth engine executes whatever profile the server publishes, a new device that speaks a similar protocol can usually be added as pure configuration — see `docs/AddingSensor.md` in the source repository for the full schema and worked examples. An app rebuild is only needed for sensors that need a transport other than Bluetooth GATT, a proprietary handshake, or a new advertisement decoder.

## 3.2 Enabling sensor sources — Integrations

The *Integrations* page (reached from the *Sensor data* tab of Survey Settings, Figure 3.2) is where a sensor type already installed on the server is switched on for the study. Each card shows one sensor type with an enable toggle, its connection timeout, and the used-data parameters it currently feeds. An *Available templates* section below lists any of the seven built-in sensors that have not been installed yet, each with a single *Activate* button that installs the type, its raw parameter catalogue and a published GATT profile in one step.

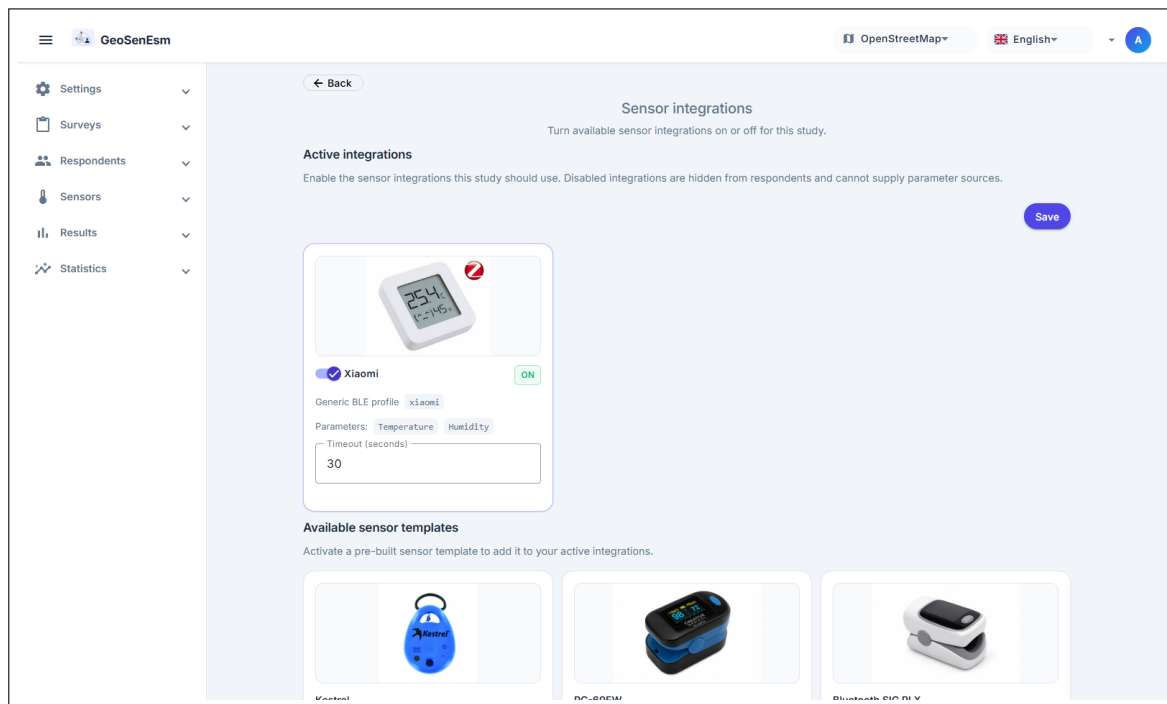


Figure 3.2: *Integrations* — enabling sensor types and activating built-in templates.

## 3.3 Adding a new sensor type — Sensor profiles

The *Sensor profiles* page (also reached from Survey Settings, Figure 3.3) is for sensors that are not one of the built-in templates. A new sensor type is created with a code, a display name and an integration mode (*profile*, for any device that fits the GATT engine, or *native*, for a sensor implemented as code inside the mobile app). Profile-mode types get a JSON editor where the discovery rule, read sequence and byte decoders are authored, validated against sample “golden packets”, saved as a draft and finally published. Every edit is versioned: the revision history lists past drafts and published revisions, with a rollback action for archived ones.

GeoSenEsm

OpenStreetMap English

Settings Surveys Respondents Sensors Results Statistics

← Back

### Add new sensor type

Create a sensor type, manage bounded GATT sequences and BLE advertisement profiles, and identify types that require a compiled mobile adapter.

New sensor type

Sensor type code Sensor type name Integration mode GATT profile engine

Create sensor type

Sensor type

Xiaomi (xiaomi) — GATT profile engine Delete sensor type

GATT profile specification

The JSON spec must match the GATT profile schema. Use Validate before publishing.

Import JSON Format

Minimum engine version

1.0.0

Profile JSON

```
{
  "discovery": {
    "nameExact": "LYWSD03MMC",
    "serviceUuid": "ebe0ccb0-7a0a-4b0c-8a1a-6ff2997da3a6"
  },
  "goldenPackets": [
    {
      "characteristicUuid": "ebe0ccc1-7a0a-4b0c-8a1a-6ff2997da3a6",
      "expected": {
```

Figure 3.3: Sensor profiles — creating a sensor type and editing its GATT profile.

## 3.4 Used sensor data and fallback sources

The *Sensor data* tab of Survey Settings (Figure 3.4) decides what actually counts as “sensor data” for the study, independently of which raw sensor types exist:

**Sensor data mode.** A checkbox that switches the whole study between “no sensor data” and “configured sensors”.

**Used parameters.** Each row is one column that will appear in exports and response documents (name, data type, unit, whether it is required). A parameter can be fed by more than one raw sensor reading — for example, temperature from both a Kestrel and a Xiaomi device — listed as an ordered list of *sources*; the first source that reports a value for a given reading wins, giving a fallback priority chain if a respondent’s primary sensor is unavailable.

**Respondent assignments.** Once at least one assignment exists, a table lists which respondent has which sensor type and physical device, with a per-assignment enable toggle and priority order. This table is the one part of sensor configuration that stays editable after the initial survey is published, since assigned hardware changes throughout a live study.



Study-wide options that apply to all surveys and respondents.

General | **Sensor data settings**

Sensor types and integrations can only be created or changed here before the initial survey is published. Once the study goes live, this setup is locked, so plan and test your sensors in advance.

**Sensor types & integrations**  
Create new sensor types and manage active integrations from here. [Add new sensor type](#) [Integrations](#)

**Sensor data configuration**  
Choose whether surveys collect sensor data and which sources can provide configured parameters.  
☒ Use configured sensors

**Sensor data columns**  
Define the parameter columns collected from sensors or manual entry. [Show inactive](#)

CURRENT PARAMETERS

| Code        | Name        | Data type | Unit |
|-------------|-------------|-----------|------|
| temperature | Temperature | Decimal   | C    |

☒ Required ☒ Active [Save parameter](#)

**Sources**

|   |                      |                   |
|---|----------------------|-------------------|
| 1 | Xiaomi (temperature) | <a href="#">×</a> |
| 2 | Manual (temperature) | <a href="#">×</a> |

| Code     | Name     | Data type | Unit |
|----------|----------|-----------|------|
| humidity | Humidity | Decimal   | %    |

Figure 3.4: Survey settings — Sensor data tab.

## 3.5 Sensor devices

The *Sensor devices* tab (Figure 3.5) is where a MAC address is tied to a sensor type and, optionally, to the respondent who carries it. This matters most for the Xiaomi LYWSD03MMC, whose Bluetooth name is identical on every unit: without a whitelist entry the mobile app cannot tell one respondent's sensor from their neighbour's. Kestrel and other devices with a unique identifier do not need a whitelist entry to be told apart, but can still be recorded here for bookkeeping.

Configure sensor MAC addresses and assign each sensor to a respondent so the mobile app can sync the correct device automatically.

[+ Add](#) [Refresh](#) [Export](#) [Import](#) [?](#)

| ID | TYPE   | MAC ADDRESS       | RESPONDENT | UPDATE            | DELETE             |
|----|--------|-------------------|------------|-------------------|--------------------|
| 1  | Xiaomi | 00:1A:2B:3C:4D:5E | 00008      | <a href="#">✎</a> | <a href="#">🗑️</a> |

Items per page: 30 1 - 1 of 1 [<](#) [>](#)

Figure 3.5: Sensor devices — MAC whitelist with respondent assignment.

## 4 Setting up survey

This chapter covers the *Surveys* and *Respondents* navigation groups: authoring a survey's structure and schedule, and managing the people who will answer it.

### 4.1 Respondents

The Respondents view lists all accounts, lets the administrator reset a password, and is where new accounts are generated (Figure 4.1).

| USERNAME | SURVEY START | SURVEY END | TIMEZONE | ID                                   | ACTIONS |
|----------|--------------|------------|----------|--------------------------------------|---------|
| 00001    | 2026-08-01   | 2026-08-31 | UTC      | a05cd40f-ad38-4cb5-a860-56630c0b27fc |         |
| 00002    | 2026-08-01   | 2026-08-31 | UTC      | e36c76dd-68c3-46f0-b7dd-b2ad58d40fb5 |         |
| 00003    | 2026-08-01   | 2026-08-31 | UTC      | aa5947d1-a64a-4553-8716-f054eb77869e |         |
| 00004    | 2026-08-01   | 2026-08-31 | UTC      | 4ba77d74-d2bb-488f-8273-124b97611db4 |         |
| 00005    | 2026-08-01   | 2026-08-31 | UTC      | ae8e8b3c-0852-475b-9786-059c3ab41c71 |         |
| 00006    | 2026-08-01   | 2026-08-31 | UTC      | bc4bd941-11db-462b-a56b-bd89e652a947 |         |
| 00007    | 2026-08-01   | 2026-08-31 | UTC      | 7c3af11c-28e8-4a3a-a4a5-e95fc15ac7d6 |         |
| 00008    | 2026-08-01   | 2026-08-31 | UTC      | 42etb8b4-2157-4dc4-97bd-7387c614e5c1 |         |

Figure 4.1: *Respondents list.*

Filters help find respondents who are not filling the questionnaires actively, including a user-name/ID search and a survey-window presence filter (Figure 4.2).

Username:  ID:  Survey window: Any Start from:  Start to:  End from:  End to:

Figure 4.2: *Advanced respondent filters.*

When a batch of accounts is created (Figure 4.3), the generated credentials are copied to the clipboard — they cannot be recovered from the server afterwards. Paste the CSV into a safe place before closing the dialog.

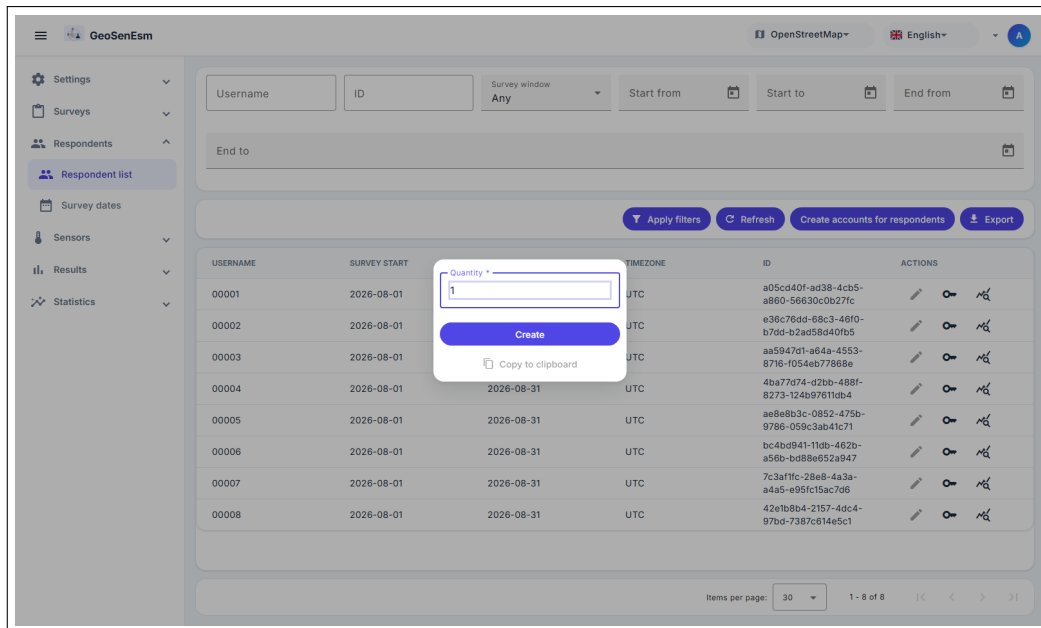


Figure 4.3: Creating respondent accounts. The generated credentials are exported via Copy to clipboard.

#### 4.1.1 Survey dates

Each respondent can optionally be assigned a *survey window* — a start and end date bounding when they are expected to participate. The *Survey dates* page (Figure 4.4) batch-assigns or clears these dates for any selection of respondents, and shows a daily activity chart of how many respondents currently have an open window. Survey windows drive the *Active* filters used in Statistics (Section 5.5), Daily completion (Section 5.5.4) and Issues (Section 5.6).

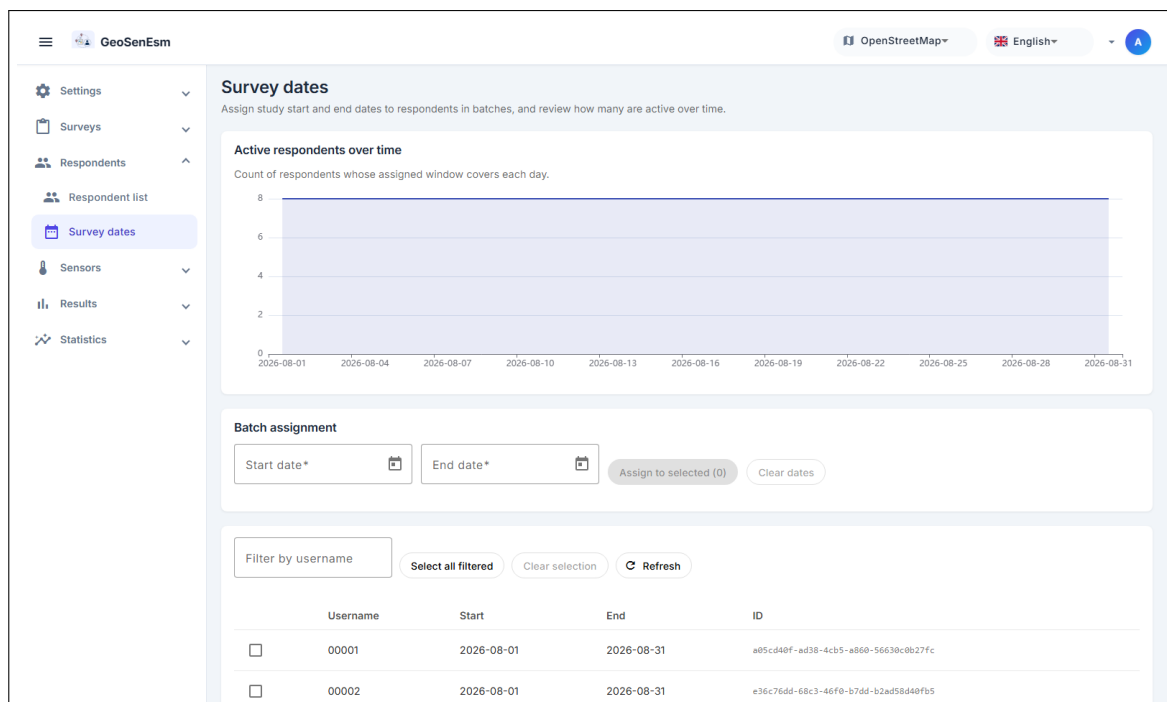


Figure 4.4: Survey dates — batch assignment and activity overview.

## 4.2 Creating a survey

Survey creation is broken into three steps: editor, sending policy (with notifications) and preview.

### 4.2.1 Editor

The editor (Figure 4.5) lets the administrator add sections and questions. Each section has one of three visibility rules (Figure 4.6a):

**Always visible.** Shown to every respondent.

**Group visibility.** Shown only to respondents whose initial-survey answers place them in a specific group.

**Conditional visibility.** Shown only if an earlier section's answer matches a rule.

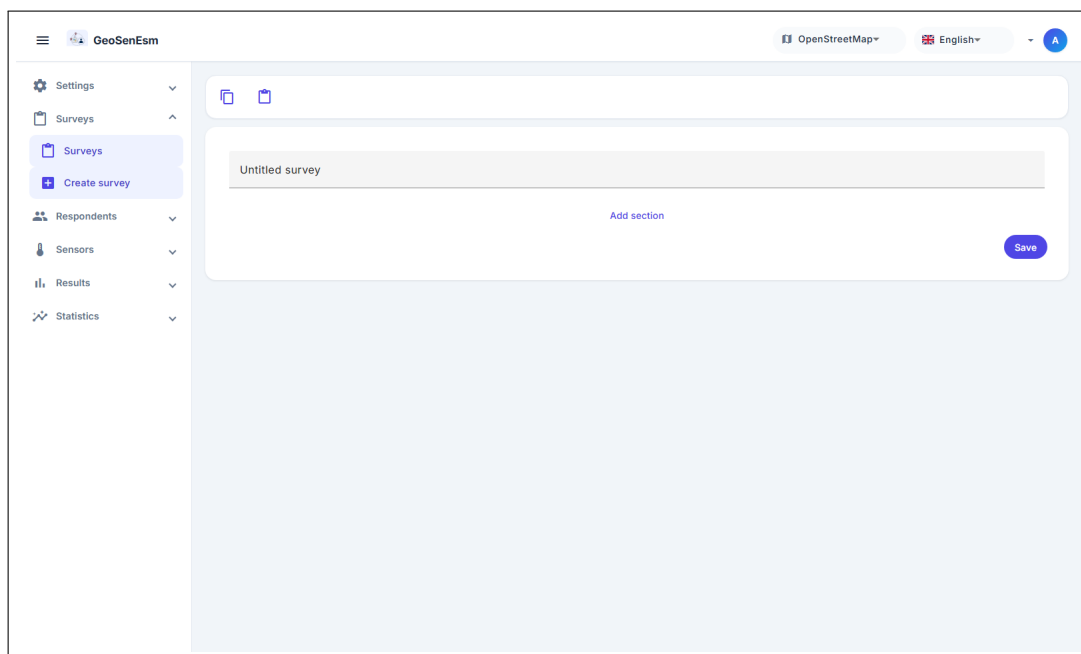
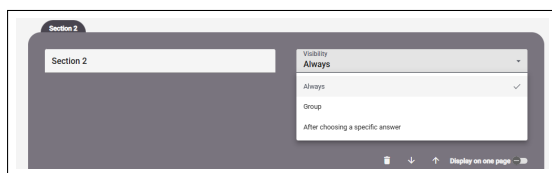
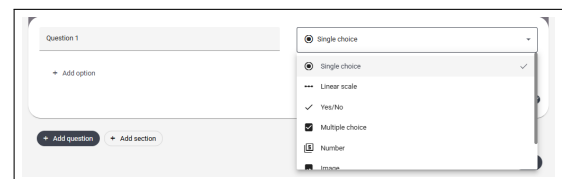


Figure 4.5: Survey editor.



(a) Section visibility rules.

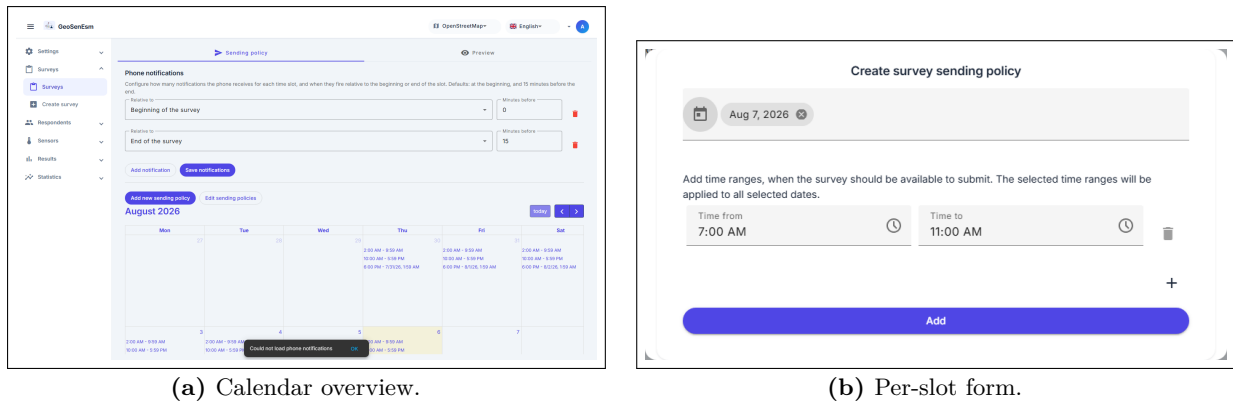


(b) Question types.

Figure 4.6: Building blocks exposed by the editor.

### 4.2.2 Sending policy and notifications

Once the structure is saved, the administrator defines the sending policy — when the survey will be open for participants. The schedule is drawn on a calendar view (Figure 4.7, left); the per-slot form is shown on the right. Every scheduled time is interpreted in each respondent's own local time zone (detected automatically by the mobile app at login), so a slot that opens at 9:00 reaches every participant at 9:00 local time regardless of where the study runs.



(a) Calendar overview.

(b) Per-slot form.

Figure 4.7: Sending-policy editor.

The *Notifications* panel (Figure 4.8) is new in v3.0.0: instead of a single fixed reminder, each survey can define up to ten notification rules, each relative to the *beginning* or the *end* of a slot and offset by a chosen number of minutes. A newly created survey starts with the previous default behaviour (a notification when the slot opens, and another 15 minutes before it closes) already filled in as editable rows.

Figure 4.8 shows the 'Phone notifications' panel. It contains a title 'Phone notifications' and a description: 'Configure how many notifications the phone receives for each time slot, and when they fire relative to the beginning or end of the slot. Defaults: at the beginning, and 15 minutes before the end.' There are two rows of configuration. The first row is 'Relative to Beginning of the survey' with a 'Minutes before' field set to '0'. The second row is 'Relative to End of the survey' with a 'Minutes before' field set to '15'. There are 'Add notification' and 'Save notifications' buttons at the bottom.

Figure 4.8: Sending policy — notification rules.

### 4.2.3 Preview and publish

The preview tab (Figure 4.9) is a final opportunity to spot mistakes. Once the survey is published, its structure becomes immutable — respondents can start filling it in, and further edits are no longer possible. The preview also offers a *Copy* action which is handy when reusing a complicated structure as the basis for a new survey.

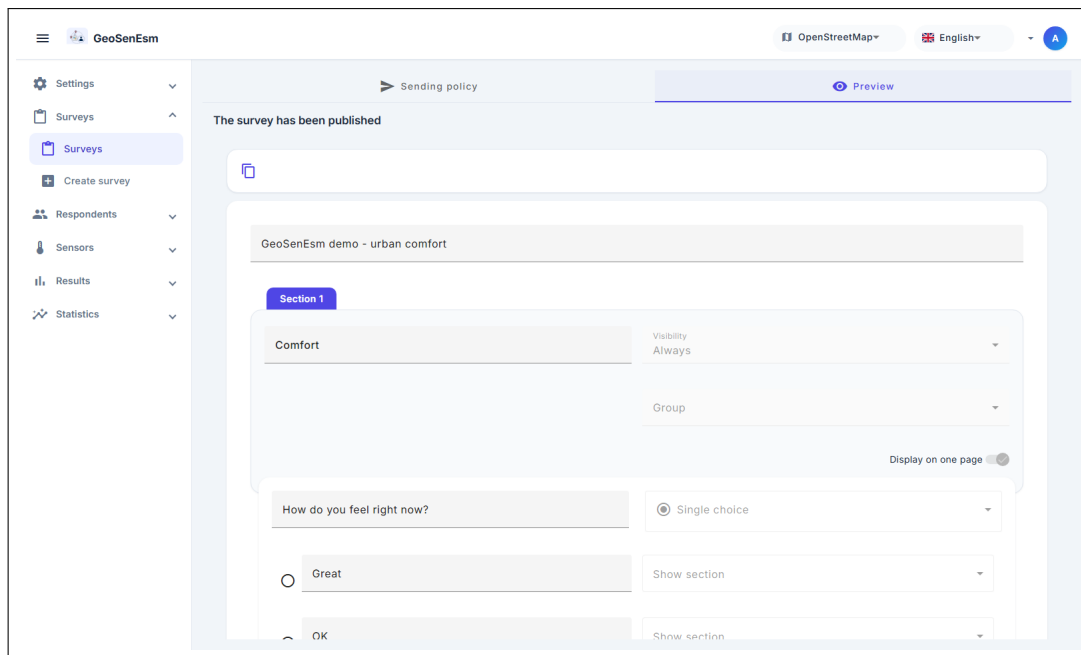


Figure 4.9: Survey preview.

## 4.3 Surveys list

The Surveys tab lists every survey together with its current sending policy. Schedules and notification rules can be edited from here at any time.

# 5 Monitoring and Survey results

This chapter covers the *Results* and *Statistics* navigation groups: everything researchers use once a study is running to inspect submitted data and spot problems early.

## 5.1 Results

The Results tab (Figure 5.1) shows a tabular view of all submitted answers. Rows can be filtered by survey and by respondent and exported to CSV, using the separators configured in Section 2.1.

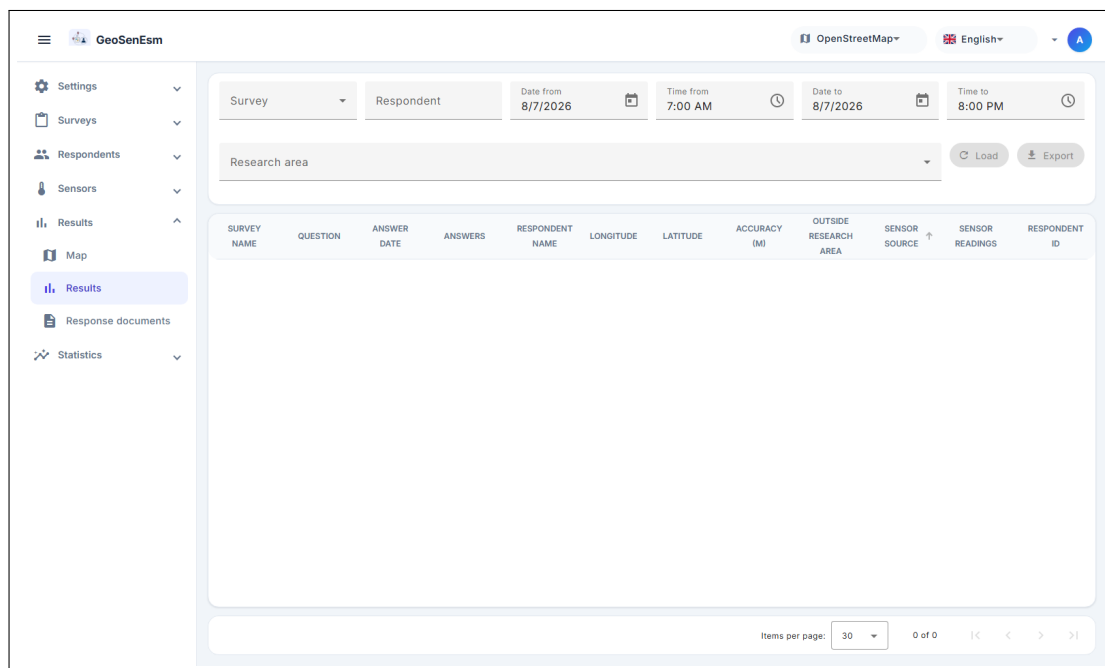


Figure 5.1: Results tab.

## 5.2 Response documents

The Response documents tab is backed by MongoDB. It contains one JSON document per submission, with the exact payload the mobile app sent to the server (answers, timestamps, embedded sensor reading, matching survey metadata). Two workflows are supported:

- **Download** one specific document from the grid by clicking the download icon in the row. The result is a self-contained JSON file that can be inspected offline.
- **Export** the full filtered set as a ZIP archive.

By default the date range is set to *today* → *tomorrow* (local time), which matches the most common use case: reviewing the responses that have just come in (Figure 5.2). The *Clear filters* button resets the range back to that default. The paginator and page-size selector stay visible even when the results table is very wide, so switching pages does not require horizontal scrolling.

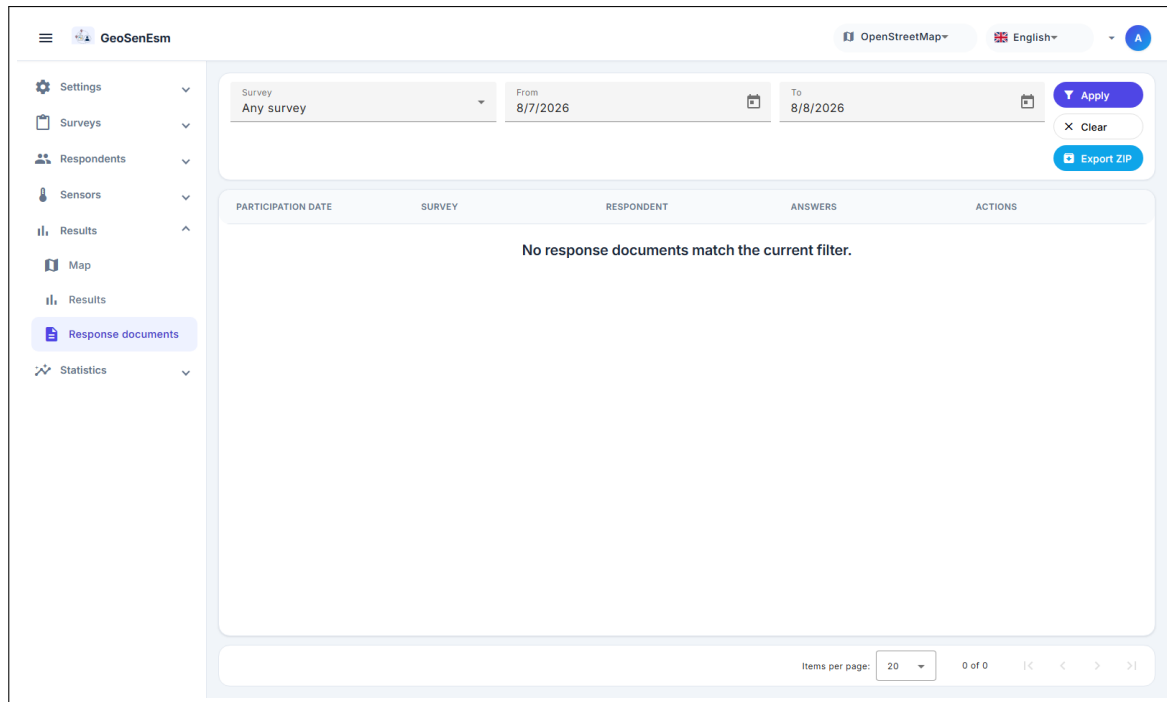


Figure 5.2: Response documents tab (Mongo-backed).

## 5.3 Sensor data

The *Sensors data* tab (Figure 5.3) lists every sensor reading uploaded by respondents. Its columns follow whichever parameters the study has configured in Section 3.4 — temperature and humidity for a study using Xiaomi or Kestrel devices, SpO<sub>2</sub> and pulse rate for one using a pulse oximeter, and so on. Filters mirror the ones available for GPS data (date range, respondent).

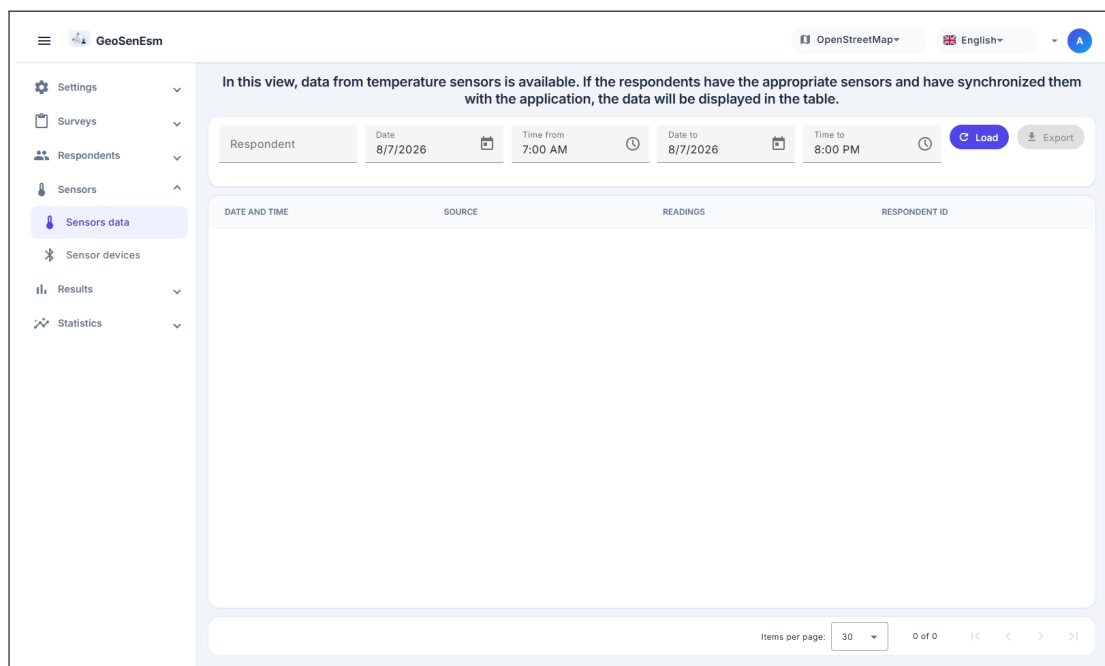


Figure 5.3: Sensor data tab.



## 5.4 Map

The Map tab (Figure 5.4) plots every GPS point in the system. Points are colour-coded by whether they lie inside or outside the research-area polygon.

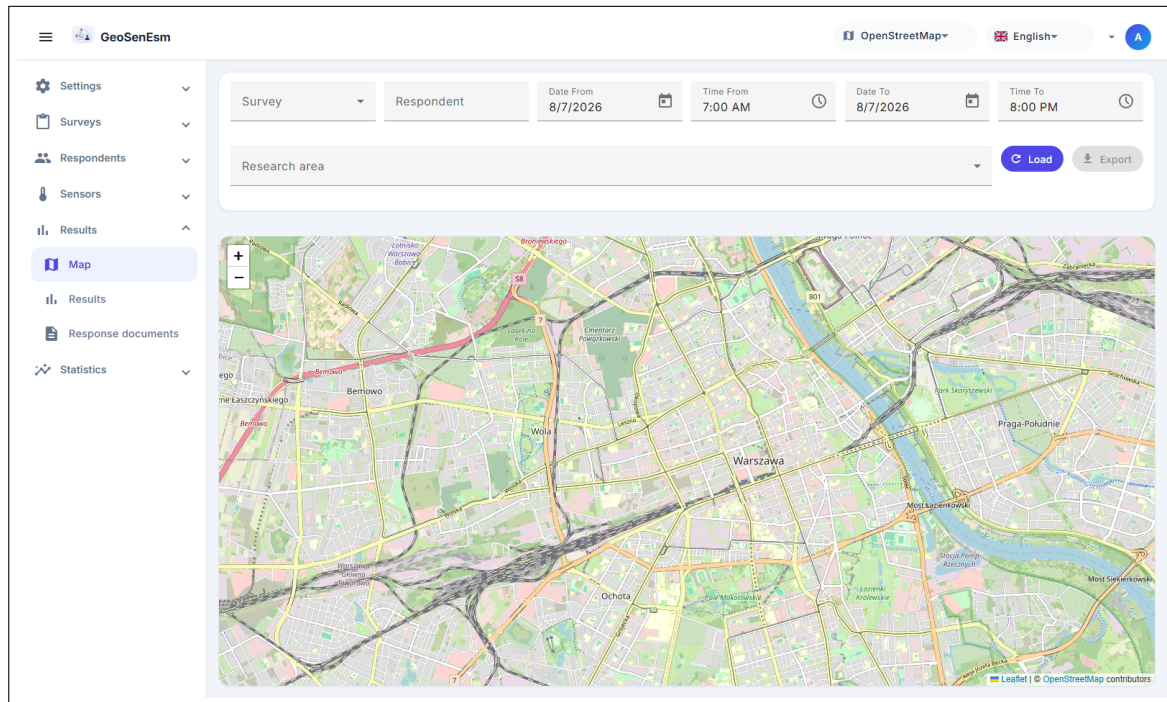


Figure 5.4: Map tab — GPS points shaded by inside vs. outside the research area.

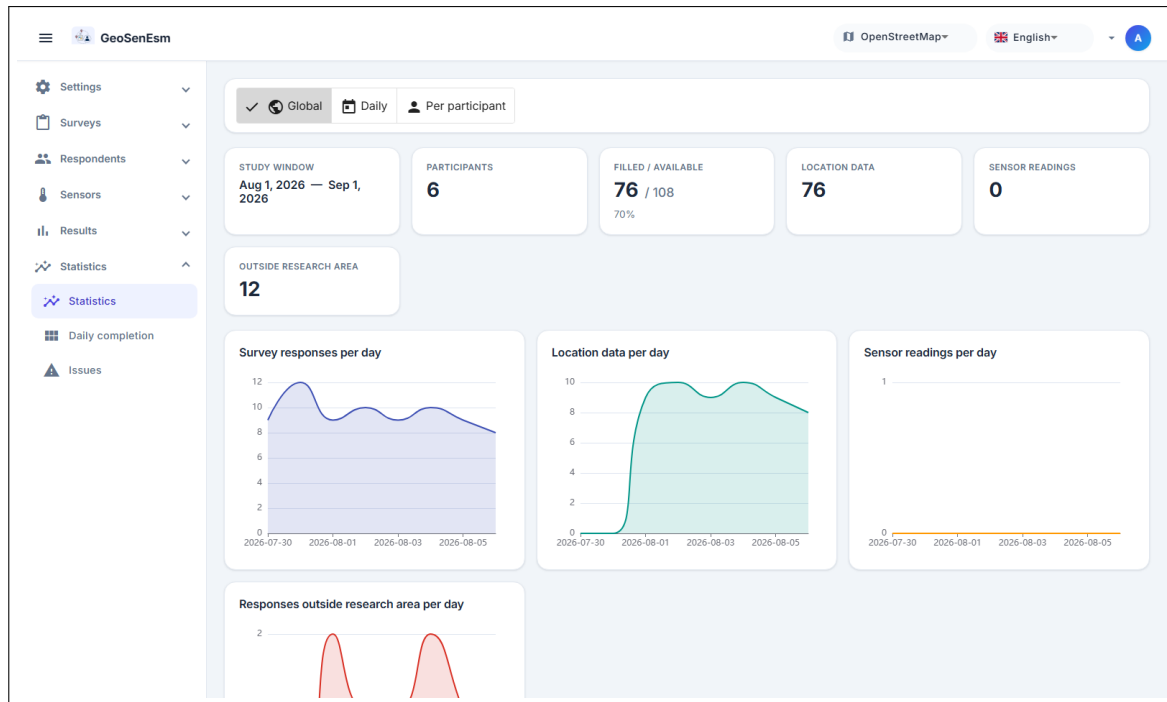
## 5.5 Statistics

The Statistics section aggregates the operational data across the whole study and is the fastest way to get a feel for how the study is going. It is split into three view-toggles at the top of the page (*Global*, *Daily*, *Per participant*) and a dedicated *Daily completion* entry in the side menu. Every view shares a common KPI strip and a set of time-series charts underneath.

Whenever data is not yet available from the backend, a spinner is shown in place of the KPIs and charts. Switching views clears the stale values so the spinner is visible during every refresh.

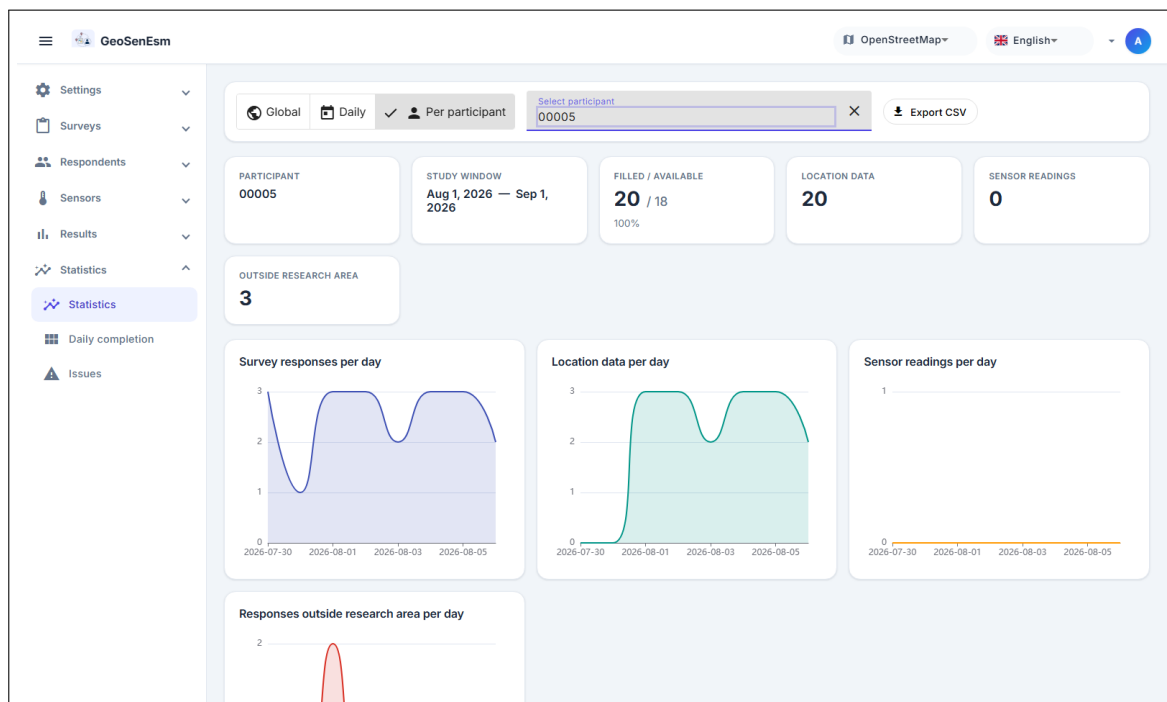
### 5.5.1 Global view

Aggregates the whole study window into per-day time series (filled surveys, GPS points, sensor readings, participations outside the research area). It is the right place to look for weekly trends or unusual dips in data collection.

Figure 5.5: *Statistics — Global view.*

### 5.5.2 Per participant view

Focuses on a single respondent. The participant is picked from a **type-to-filter** input — type a substring of a username and the dropdown narrows to the matching accounts. Once a participant is chosen, the KPI strip and per-day time series update accordingly (Figure 5.6).

Figure 5.6: *Statistics — Per participant view with username autocomplete.*

### 5.5.3 Daily view

Zooms into a single calendar day, interpreted in each respondent’s own local time. Instead of per-day series, the charts show per-hour buckets, which is useful when a slot schedule places several surveys inside the same day (Figure 5.7). The daily view also exposes an extra KPI, *Filled / Available active*, which counts only respondents that have submitted at least one survey inside the trailing 3-day window ending at the selected day. Hovering the label opens a tooltip that explains the definition.

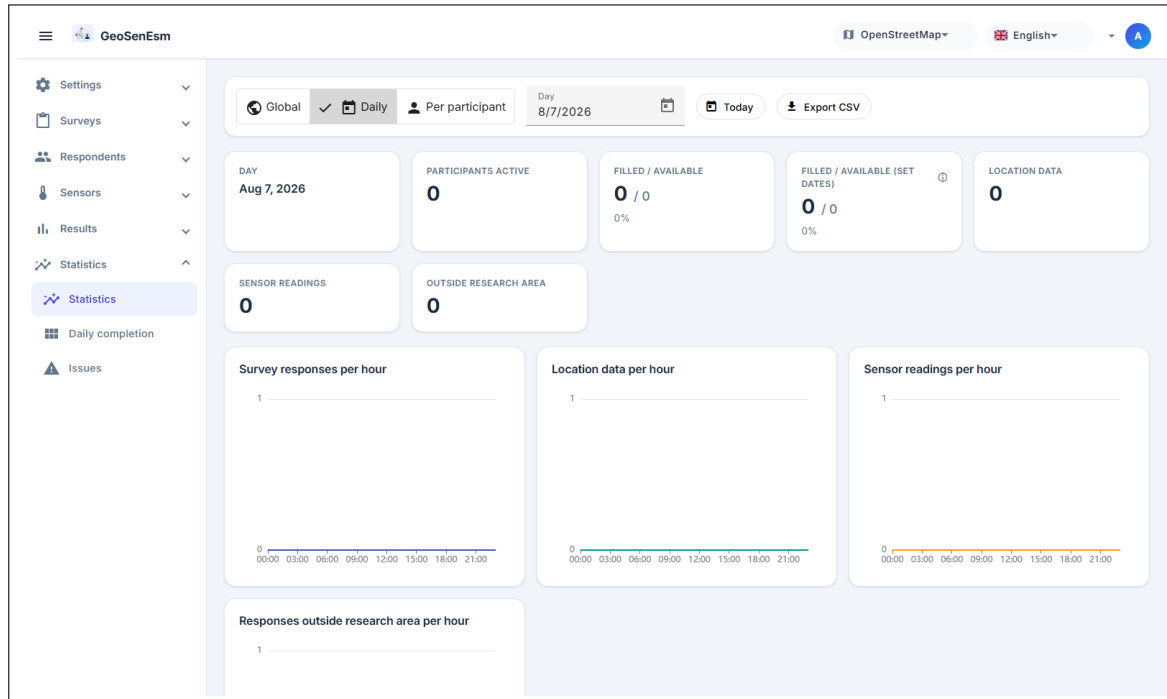


Figure 5.7: *Statistics — Daily view with hourly buckets and the “Active” KPI.*

### 5.5.4 Daily completion

Daily completion (Figure 5.8) is the compliance view. For a chosen day it draws one small card per respondent with a coloured stripe for every scheduled slot. The colour encodes what has been collected against that slot:

#### Daily-completion colour legend

- Submission collected with GPS *and* sensor data.
- Submission collected with GPS only.
- Submission collected with sensor only.
- Submission collected but neither GPS nor sensor data was linked.
- Slot has finished and no submission arrived.
- Slot has not started yet.

Each participant card can be resized: the toolbar exposes two sliders (width 110–220 px, height 90–160 px, step 5 px) with the selected size persisted in `localStorage` so the layout is stable across sessions.

The toolbar additionally offers a *Only active* filter with an adjustable window (default 7 days), see Figure 5.9. When checked, the grid hides respondents whose last submission is older than  $N$  days, measured *relative to the day chosen in the calendar* — i.e., the window is [end-of-day –  $N$  days, end-of-day], so inspecting a past day never counts submissions that arrived after it.

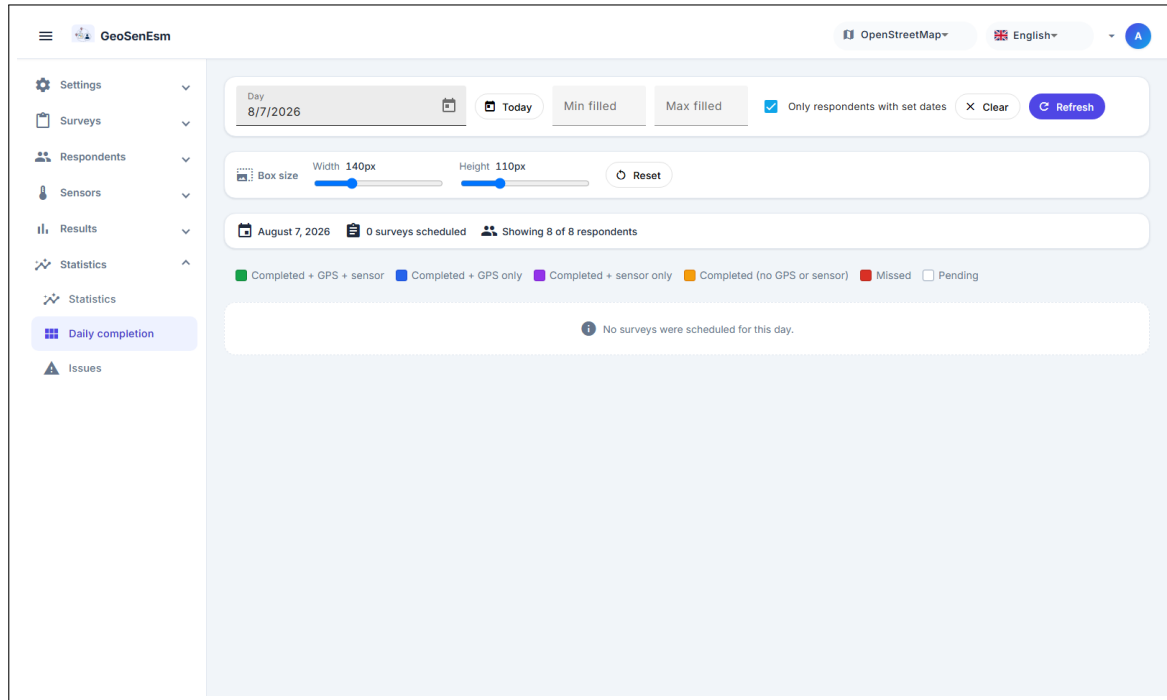


Figure 5.8: Statistics — Daily completion.

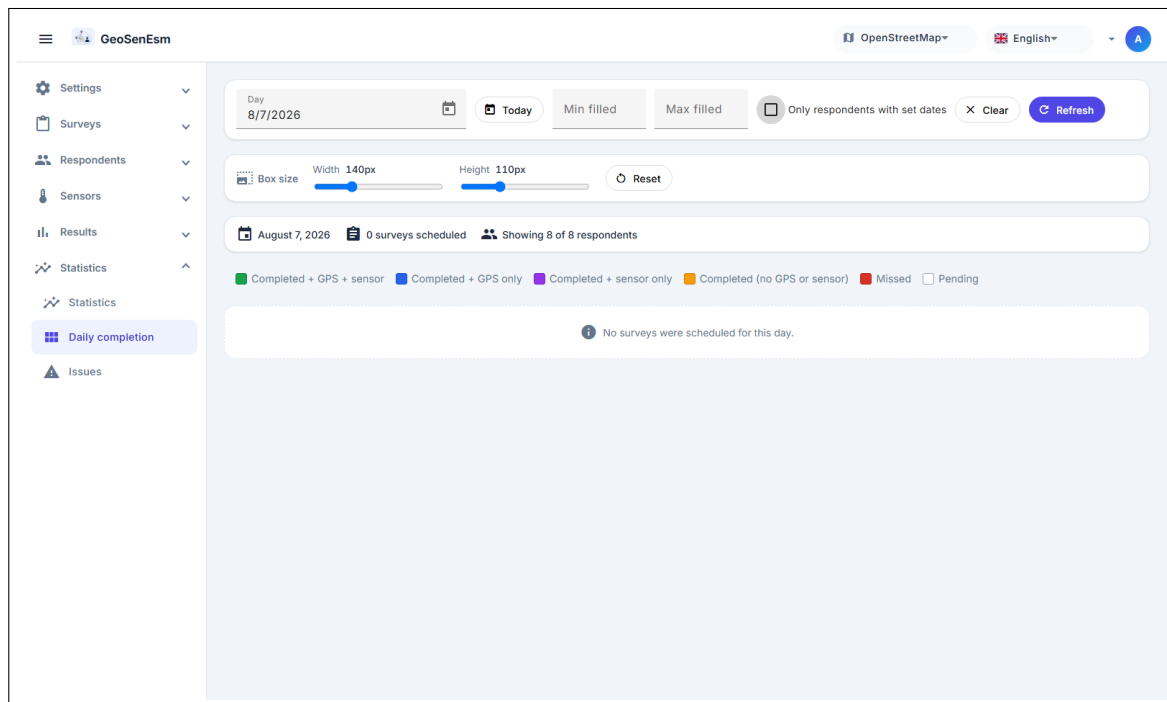


Figure 5.9: Daily completion with the “Only active in last  $N$  days” filter engaged — inactive respondents disappear from the grid.

### 5.5.5 Surveys outside research area

When a research-area polygon is defined, every GPS point that arrives is compared against it. Statistics includes a dedicated KPI card and time series that count submissions whose linked GPS row falls outside the polygon (Figure 5.10). Present on the Global, Daily and Per participant views, this number is a quick sanity check that respondents actually stayed within the study area.

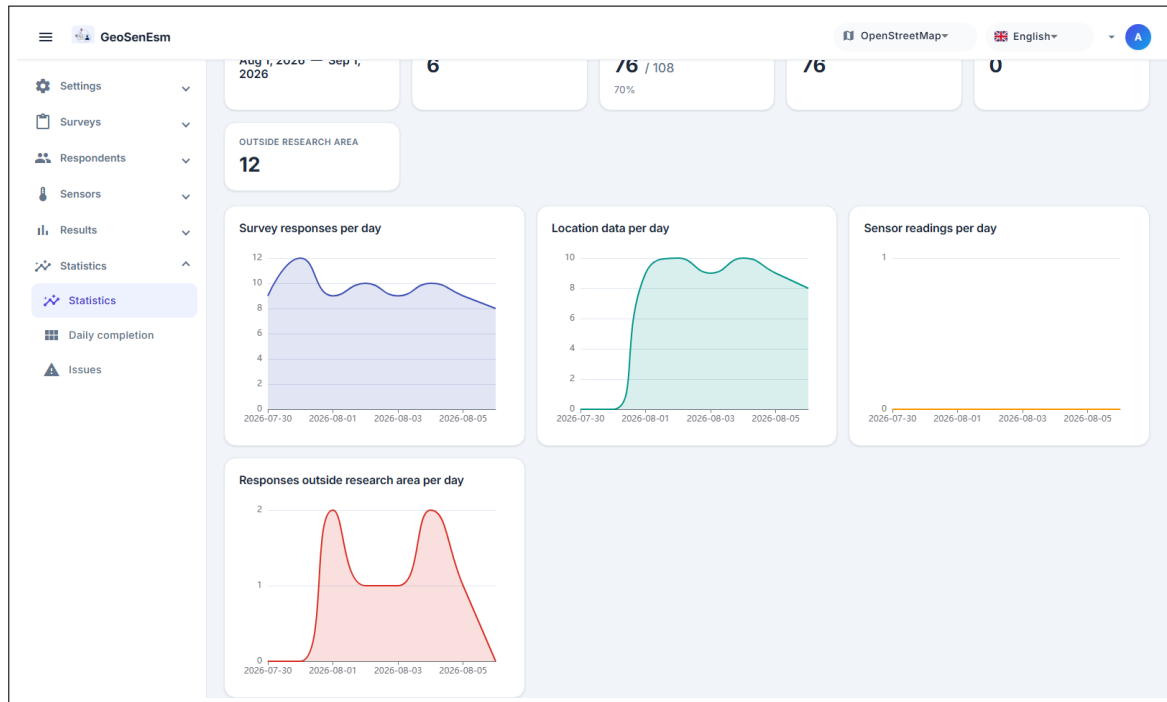


Figure 5.10: “Surveys filled outside research area” KPI and chart.

## 5.6 Issues

The Issues page (Figure 5.11) is new in v3.0.0 and answers a single question per respondent: *are they actually completing the study?* For each respondent with a survey window assigned (Section 4.1.1) — or for a custom date range — it computes the share of scheduled slots that were filled, the share with a linked GPS point, and the share with a linked sensor reading. Numeric filters (minimum skipped count, maximum survey/GPS/sensor fulfilment percentage) narrow the sortable, paginated table down to the respondents who most likely need a follow-up message or a hardware check.

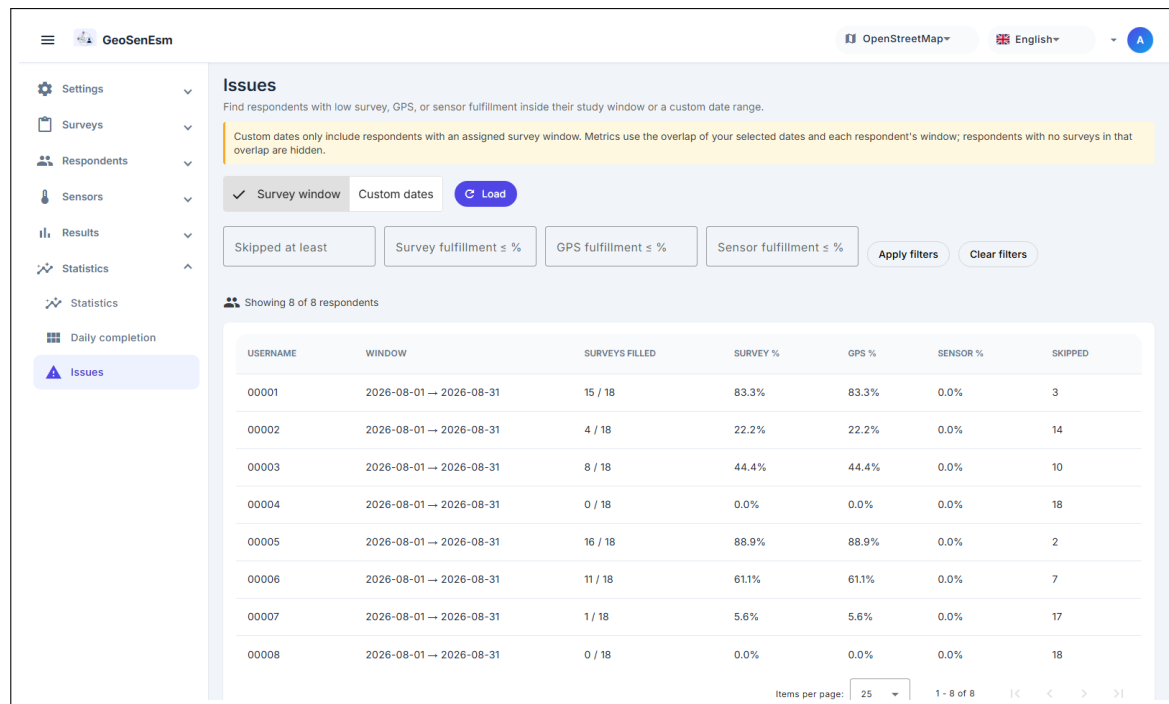


Figure 5.11: *Issues* — per-respondent survey, GPS and sensor fulfilment.

# 6 Mobile application

## 6.1 Login and first-run setup

The first screen the user sees is the login form (Figure 6.1). Respondents fill in the server URL (*API URL*, provided by the study administrator) together with their *username* and *password*. Accounts are created centrally in the admin panel and distributed to respondents out of band (email, print-out, and so on); self-registration is not possible. On login, the app also detects the device’s time zone and reports it to the server, so every subsequent slot and notification time is computed in the respondent’s actual local time (Section 4.2.2).

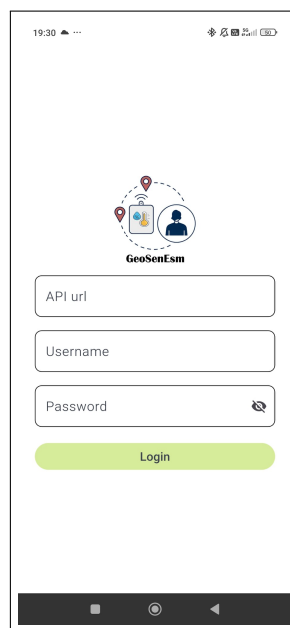


Figure 6.1: Login screen of the mobile app.

On the first launch the user is presented with the privacy policy and, for studies still using the legacy manual choice, picks their sensor type (none, Xiaomi or Kestrel). Whenever the administrator has assigned a device configured through the sensor-profile engine (Section 3.3), the app instead syncs that assignment automatically in the background and shows it as “Assigned by administrator” — no manual pairing step is needed. Then the *initial survey* is shown (Figure 6.2); its answers place the respondent in a group that later surveys can target through section-level visibility rules.

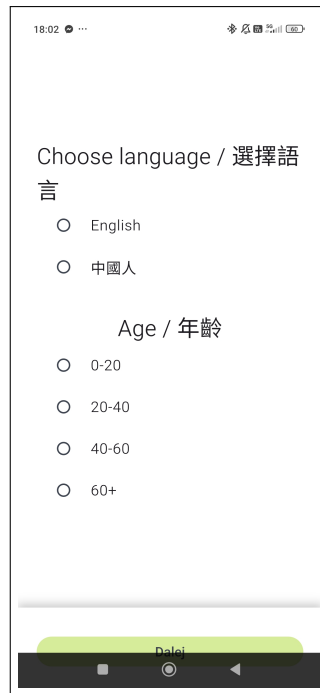


Figure 6.2: Initial-survey example (language and age group).

The app then requests the runtime permissions it needs. All three requests happen on first launch and are shown side by side in Figure 6.3.



Figure 6.3: First-run permission requests.

## 6.2 Home screen

The home screen (Figure 6.4) is intentionally minimalist. Bright orange buttons show the surveys that are currently open, together with a countdown to the slot's finish time. Navigation to the menu and to the user profile lives in the top-left and top-right corners. The layout respects the device's font-size accessibility setting.



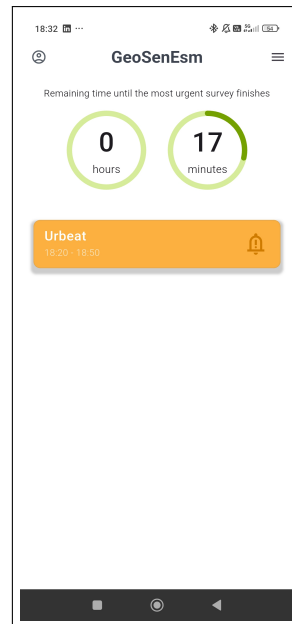
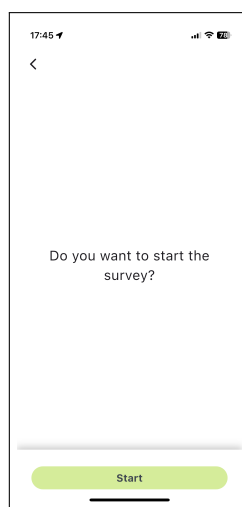


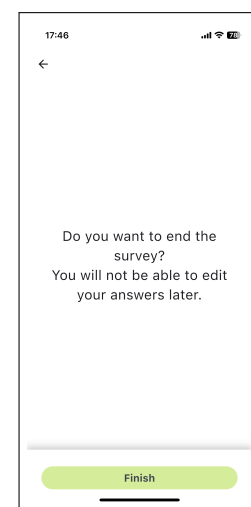
Figure 6.4: Home screen of the mobile app.

### 6.3 Filling in a survey

1. **Open the app** and tap the orange button whose caption matches the active survey. The countdown next to the button shows how long the slot is still open. Bluetooth should be enabled: the app tries to read a fresh reading from the assigned sensor while the survey is being filled in.
2. **Introduction and outro screens** (Figure 6.5) confirm the start of the survey and the submission of responses. Once submitted, a response cannot be edited.
3. If the respondent's study has no sensor configured, or the respondent chose *no sensor* on the first-run setup, the app additionally asks whether they want to proceed without a sensor reading (Figure 6.6).



(a) Introduction.



(b) Submission confirmation.

Figure 6.5: Confirmation screens shown before and after a survey.

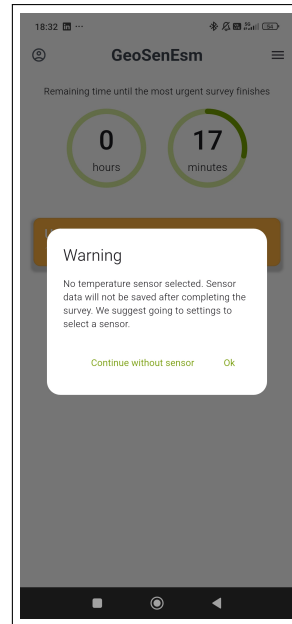
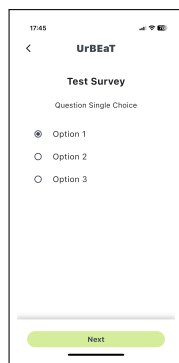


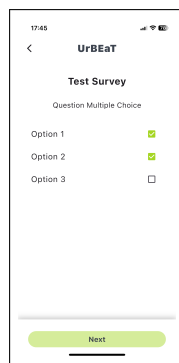
Figure 6.6: “No sensor” warning shown before filling a survey.

## 6.4 Question types

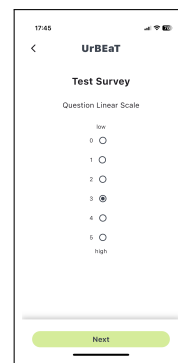
Surveys support seven question types (Figure 6.7).



(a) Single choice.



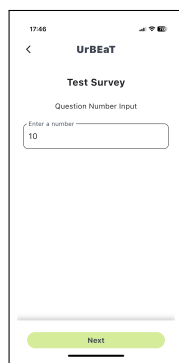
(b) Multiple choice.



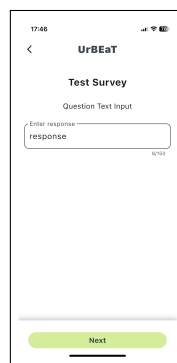
(c) Linear scale.



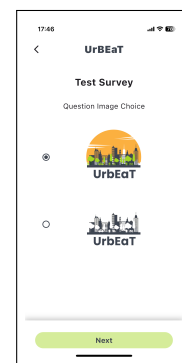
(d) Yes / No.



(e) Number input.



(f) Text input.



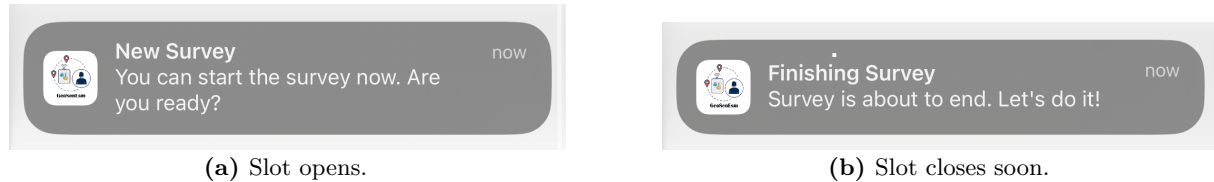
(g) Image choice.

Figure 6.7: The seven question types supported by GeoSenEsm.

On submission the app sends the response to the server together with the matching location and sensor reading.

## 6.5 Notifications

Respondents are notified according to the notification rules the administrator configured for that survey (Section 4.2.2); by default, that means once when a slot opens and again 15 minutes before it closes (Figure 6.8), computed in the respondent's own local time.



(a) Slot opens.

(b) Slot closes soon.

Figure 6.8: Push notifications sent by the mobile app.

## 6.6 Offline mode

The app works both online and offline. Once a device has synced with the server, all published surveys are cached locally. Respondents can complete them without a network connection; answers, GPS points and sensor readings are pushed to the server as soon as connectivity returns.

## 6.7 Additional views

GeoSenEsm exposes what the app has captured to the respondents themselves, building trust and giving them a way to spot problems (for example, a Bluetooth issue that stopped sensor readings for a week).

### 6.7.1 Map

The map view (Figure 6.9, left) plots every submitted location point. Tapping a marker opens its details (right).

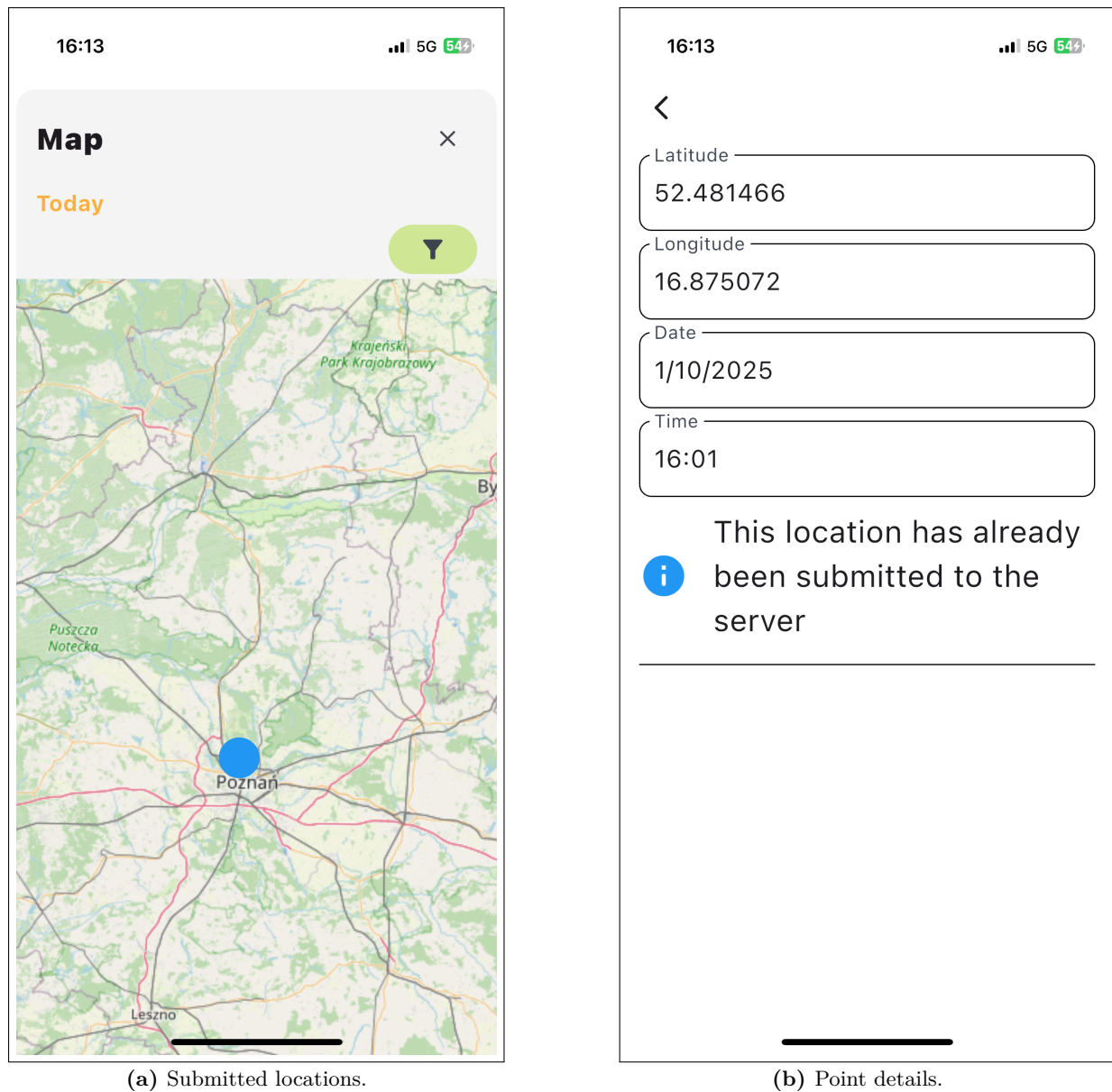
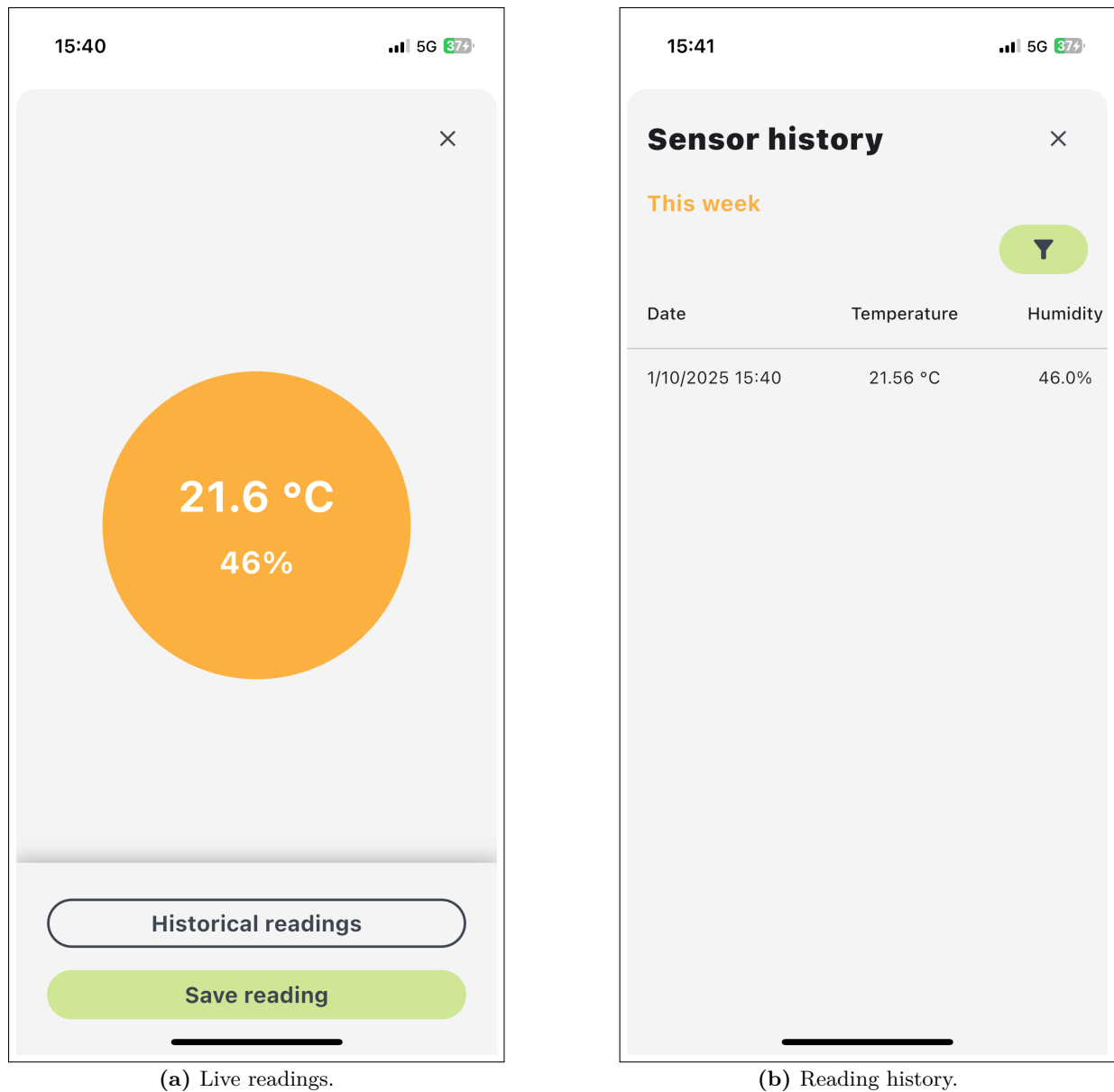


Figure 6.9: Mobile map module.

### 6.7.2 Sensor data and history

The sensor module shows live readings; the history sub-view lists every reading the app has captured (Figure 6.10). Both views are driven by whichever parameters the assigned sensor profile reports — typically temperature and humidity, but SpO<sub>2</sub> and pulse rate for a pulse oximeter, or moisture and light for a plant sensor, without any change to the app itself.

Figure 6.10: *Sensor modules on the mobile app.*

### 6.7.3 Progress calendar

The calendar (Figure 6.11) colour-codes the respondent's participation history:

- ■ Green — surveys completed on time.
- ■ Red — surveys missed.
- ■ Yellow — surveys still upcoming.

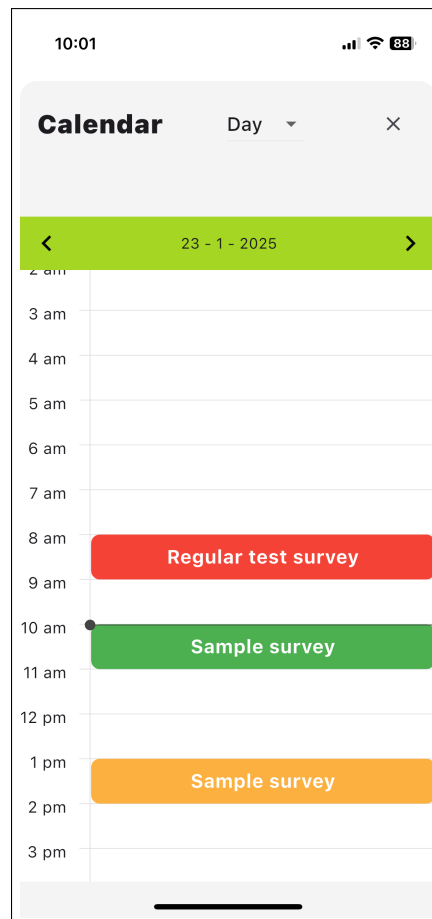


Figure 6.11: Colour-coded participation calendar.

## 6.8 Design consistency

Surveys deliberately avoid complex gestures such as swiping or drag-and-drop. One question is shown per page and can be answered without scrolling. The colour palette (orange, green, dark grey) and the study logo are used consistently across the mobile app and the admin panel.

## 7 Closing notes

The design of GeoSenEsm follows a few consistent principles:

- *Prioritise the respondent.* Keep the mobile UI simple, the permissions transparent and the sensitive data controllable.
- *Meet the researcher where they are.* Every artefact (results, response documents, GPS, sensor readings) is exportable to CSV, JSON or ZIP so it can flow straight into existing analysis pipelines, in the researcher's own CSV convention.
- *Configuration over code.* Adding a new Bluetooth sensor, tuning a notification schedule, or changing what counts as "sensor data" are all admin-panel actions, not mobile-app releases.
- *Make problems easy to spot.* The Statistics dashboard, the Daily completion grid, the Issues page and the outside-research-area KPI are all there so an issue with a single respondent or a single day of the study is visible at a glance.

The Experience Sampling Method integrates naturally with a mobile application: participants carry their phone with them at all times, and the app can capture context (location, environmental or biometric sensor readings) alongside their subjective response. Cross-referencing the two lets researchers build a richer picture of how urban environments and physiological state shape everyday behaviour.